An hourly kilometre-scale constrained reconstruction of fine particulate matter for Kandy, Sri Lanka: construction, evaluation under sparse local monitoring, and measurement priorities

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The problem

Fine particulate matter (PM2.5) carries the largest estimated burden of disease of any air pollutant, and monitoring is sparsest where concentrations are highest. Kandy is such a city: a valley settlement in the central highlands of Sri Lanka with 98,828 residents inside its municipal boundary (2012 census) and nearly 389,000 commuters entering on a typical weekday (World Bank, 2020), so that most of those exposed to its air during the day do not live in it. It has no reference-grade PM2.5 monitor in operation and two low-cost sensors with public records. Any PM2.5 field for Kandy is therefore a model output that almost no local observation can evaluate.

Aim

To reconstruct PM2.5 over the Kandy basin at hourly and one-kilometre resolution for 2019 to 2023 from data available everywhere; to establish how far each part of that reconstruction can be trusted; to determine the spatial scale below which it is no longer supported; and to identify the measurements that would most improve it.

The model

Concentration on the kilometre grid is a regional background, uniform across the city at any hour, plus a local increment redistributed by a pattern of unit spatial mean:

$$\mathrm{PM}(x,y,t) = B(t) + \max(\Delta,0)\,P(x,y,t) + \min(\Delta,0) + \varepsilon(t)\,[P(x,y,t)-1], \qquad \Delta(t) = T(t) - B(t)$$

T, the basin-mean concentration, is a gradient-boosted model of the residual between the two local sensors and a composition-model prior, driven by reanalysis meteorology and satellite retrievals, shifted each year to a satellite-derived area mean and shaped to the sensors' daily and seasonal cycles. B is a regional-background estimate built from the same satellite level, the composition model and an air-mass classification, capped so that it never reaches the hourly total. P combines a satellite surface with a terrain-confinement term; a terrain-steered dispersion step was removed after it lowered agreement with held-out stations. The field averages to T (to within 0.0 per cent in practice), so misplacing material cannot create more of it, and each component declares which observations it may use. The local increment is a residual of this decomposition, not the mass emitted inside the city.

Validation without local ground truth

The two Kandy sensors trained and shaped T, so they cannot validate it. Three other lines of evidence are used, and their results are graded rather than combined. The complete model was built in ten monitored valley and basin cities with only two anchor stations, as at Kandy, and scored against withheld stations: the seasonal cycle carries everywhere (0.943 to 0.996) and the daily cycle unevenly (-0.325 to 0.977), both about as well as the two anchor stations' own climatology. The deployed temporal model was placed on a cross-city ladder of information budgets in 104 cities (exploratory, specified before scoring): where its anchor stations observed the period it cut daily error by 32.7 per cent against a generic sensorless estimate (correlation 0.864); where they did not, by 4.83 per cent, not distinguishable from zero. Its satellite-derived level sat 7.87 per cent above the withheld city means and its nominal 90 per cent interval covered a fraction 0.696. At Kandy the national research organisation's daily record for 2021 and 2022, recovered from a published figure, matches the field's seasonal shape (correlation 0.767 across calendar

months) but not its day-to-day variation (within-month correlation 0.289), and the field misses the crisis year 2022. Trained on one Kandy sensor, the temporal model tracks the other at a daily correlation of 0.521 and 0.615. Against the regulatory station in Kandy city, whose public record covers 858 days of 2019 to 2023, the field's daily correlation is 0.6 (0.73 within the month).

The reconstructed field, and how far it can be trusted

The modelled annual basin mean is 13.6 to 16.8 micrograms per cubic metre over 2019 to 2023, above the World Health Organization guideline of 5 in every year. Taken from the satellite surface the level read 24 per cent above the regulatory station; the corrected low-cost record is within -5 per cent of the station, so the excess lies in the satellite level. The delivered level is re-anchored to the station by a factor of 0.8 fitted on 2019 to mid-2021; on the held-out 2022 and 2023 days it is within -1 per cent of the station (satellite level 24 per cent), over only 11 months. The exceedance is robust: the annual means stay at least 2.4 times the guideline across the factor's interval. The seasonal cycle is the best-supported property, with the maximum in the north-east monsoon (24.3 in December to February of 2023) and the minimum in the south-west monsoon (9.3). The day-to-day sequence is supported on days the sensors reported. Morning and evening peaks either side of a midday dip are robust in timing, but their depth depends on the humidity correction of the sensor record: with hourly rather than constant humidity the morning-peak-to-midday ratio falls from 1.93 to 1.68 and the night falls below midday, while the level, the daily sequence and the local fraction barely move. The regulatory station's hourly record has the same timing, but its deep night is the daily minimum (0.71 of midday), so the delivered cycle was refitted to the station on 2019 to mid-2021; on later hours its night runs 0.77 of midday, against 1.17 before. The interval, too narrow there at first (68.7 per cent of held-out hours), was widened on the same early years and now contains 87.1 per cent. Under the baseline decomposition the local increment is 0.505 of the modelled total, 0.447 to 0.518 across the tested ways of applying the cap and 0.565 with a 48-hour background window: a property of the decomposition, not a source apportionment. At the regulatory station the local share has a lower bound of 0.36 and a tracer estimate of 0.53; the model may be high.

What each additional observation is worth

Because every component declares its inputs, an observation can be withheld and the loss measured across monitored cities. A registered confirmation on 72 fresh cities found two stations used only to recalibrate a free estimate worth 8.53 per cent and a background series read daily worth 41.1 per cent. A later review showed the comparison was not like for like. In a post hoc re-analysis that used every source on the day, one station read daily reduced daily error by 42.1 per cent, two by 53.3 and five by 58.4, against about 11.0 per cent for stations used only as a calibration, and no ordering of station types was established. For Kandy the priority is continuously reporting stations whose readings reach the estimate every day, with one of reference grade to settle the level and the humidity correction.

Limitations

The field is not supported below the kilometre scale. Two sites three hundred metres apart inside one model cell differ in measured PM10 by a factor of 27.5, though on different days, so the pair illustrates sub-grid variation without isolating location; on densely monitored cities the spread inside a cell (1.27) exceeds the spread between cells (1.05). A dispersion step meant to place the local increment ranked held-out stations at 0.274, below the undispersed surface (0.371) and a free built-up layer (0.393), and was removed; three to eight stations do not rank neighbourhoods usefully either. The map is a hypothesis for testing, not a ranking of neighbourhoods. The chemical check is a consistency check within a composition model, not a chemical measurement at Kandy.

Ongoing work and verification

The research that follows treats the delivered field as a locked hypothesis and collects observations able to refute it: the regulatory record's provenance; sensors tested against the reference over two monsoons; a network whose level, within-cell, drainage and daily-cycle tests are powered; and a rural background reference. Every value is regenerated from source at each build by a gate that refuses to build when prose and data disagree (github.com/daminda1108/kandy_pm2.5_framework); twenty-three of 105 registered predictions were refuted and are reported as such.

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